REAL ESTATE SALES PERFORMANCE AND CUSTOMER INVESTMENT BEHAVIOUR ANALYSIS IN NIGERIA

Project Scope: Explore, analyze, and visualize real estate sales data, uncover patterns, evaluate performance, and provide actionable business insights.

Dataset/Date: 500 land transactions across different regions in Nigeria dated from 1st of January 2024 to 31st of December 2024.

EXECUTIVE SUMMARY

This project analyzes a dataset containing 500 land transactions across different regions of Nigeria which are North East, North West, North central, South East, South West, South South. The dataset contains information on property types, land size, price per square meter, total price, revenue, customer demographics, payment methods, real estate companies, marketing channels, transaction dates, and customer satisfaction.

Key findings:

- Total gross transaction value (Total Price): ₦8,138,201,242.
- Total net revenue (after discounts): ₦7,738,287,556.
- Average transaction revenue is ₦15,476,575.
- Average land size is 603.88 SQM | Average price per SQM is ₦26,923.
- North East is the region with the highest revenue with ₦1,479,820,169.
- MetroLand Developers is the best performing company with ₦1,757,314,263.
- Commercial Land in Property type contributes the largest share of revenue with ₦2,667,382,613.
- Referral is the most effective marketing channel with revenue of ₦2,319,461,640.
- First-time Buyers generate the highest revenue among customer segments with revenue of ₦2,780,486,684.
- Installment payments outpace Outright payments in revenue contribution.

OBJECTIVE

The main objective of this analysis is to explore and visualize real estate sales data in order to:

- Evaluate sales and revenue performance.
- Identify regional and property level sales patterns.
- Understand customer purchasing behavior.
- Analyze payment and marketing channels.
- Examine the relationship between land size and price.
- Identify high-value transactions and customer segments.
- Analyze monthly sales and revenue trends.
- Provide useful insights for real estate business decision making.

EXPLORATORY DATA ANALYSIS

- **Distribution of Transactions Across Regions**

Regional Transactions

North East	98
South East	95
South West	85
North Central	80
North West	72
South South	70

With total transactions of 500, North East recorded the highest number with 98 transactions, while South South recorded the lowest with 70 transactions.

- **Distribution of Property Types**

Property Type	Transactions
Commercial Land	173
Mixed-use Land	164
Residential Land	163

With a total of 500 transaction count ID, Commercial Land has the highest number of transactions, accounting for 173 transactions.

- **Distribution of Customer Types**

Customer Type	Transactions
Investor	172
First-time Buyer	168
Developer	160

With a total transaction of 500, Investors represented the largest customer group with the highest transaction, although the three customer categories were relatively close in transaction volume.

- **Payment Type Distribution**

Payment Type	Transactions
Outright	254
Installment	246

Outright payment was slightly more common than installment payment.

- **Transactions by Real Estate Company**

Real Estate Company	Transactions
MetroLand Developers	114
Greenfield Estates	113
GoldenPlots Nigeria	93
PrimeLand Ltd	93
UrbanEdge Realty	87

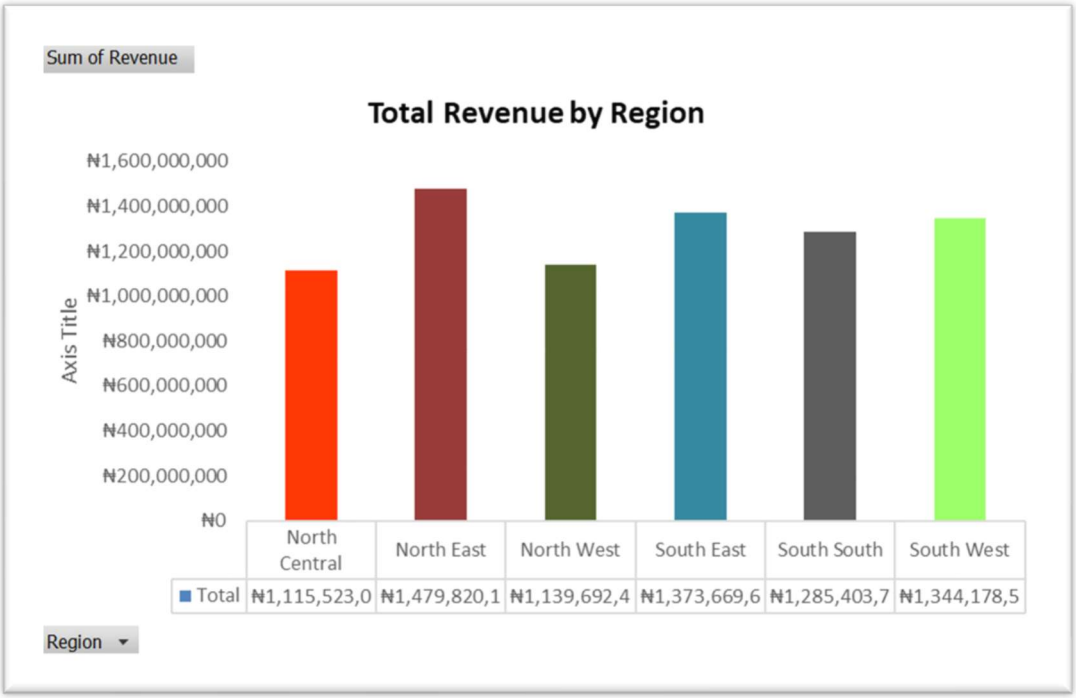
With a total of 500 transactions, MetroLand Developers recorded the highest number of transactions.

DATA SUMMARISATION

Revenue by Region

Region	Sum of Revenue
North Central	₦1,115,523,010
North East	₦1,479,820,169
North West	₦1,139,692,482
South East	₦1,373,669,679
South South	₦1,285,403,701
South West	₦1,344,178,514
Grand Total	₦7,738,287,556

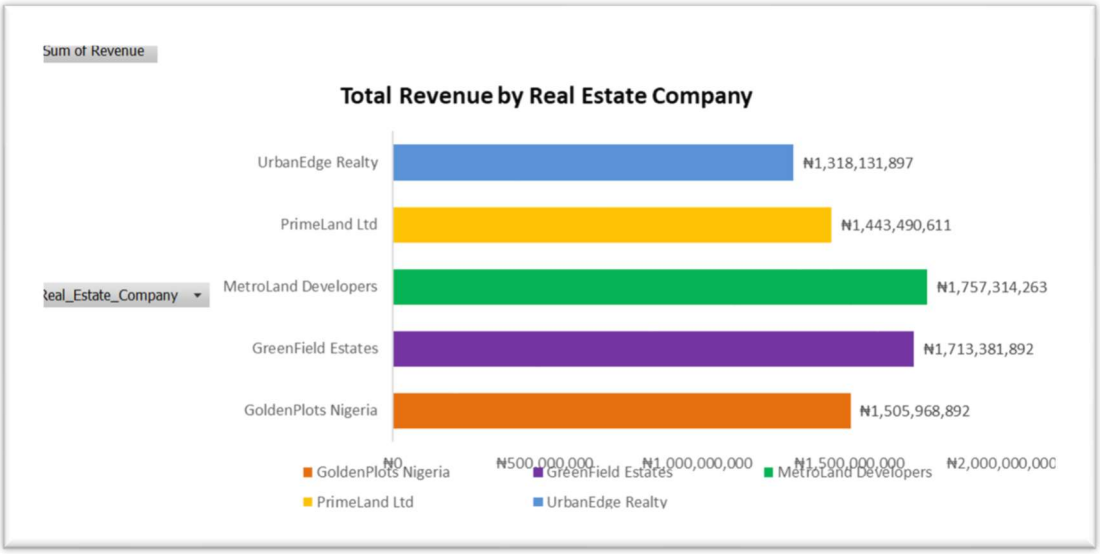
The North East generated the highest revenue while North Central has the lowest revenue



Revenue by Real Estate Company

Real Estate Company	Sum of Revenue
GoldenPlots Nigeria	₦1,505,968,892
GreenField Estates	₦1,713,381,892
MetroLand Developers	₦1,757,314,263
PrimeLand Ltd	₦1,443,490,611
UrbanEdge Realty	₦1,318,131,897
Grand Total	₦7,738,287,556

MetroLand Developers ranks first in revenue, with GreenField Estates a close second. The gap between the top two and the remaining three companies is modest, indicating a competitive market.



Revenue by Property Type

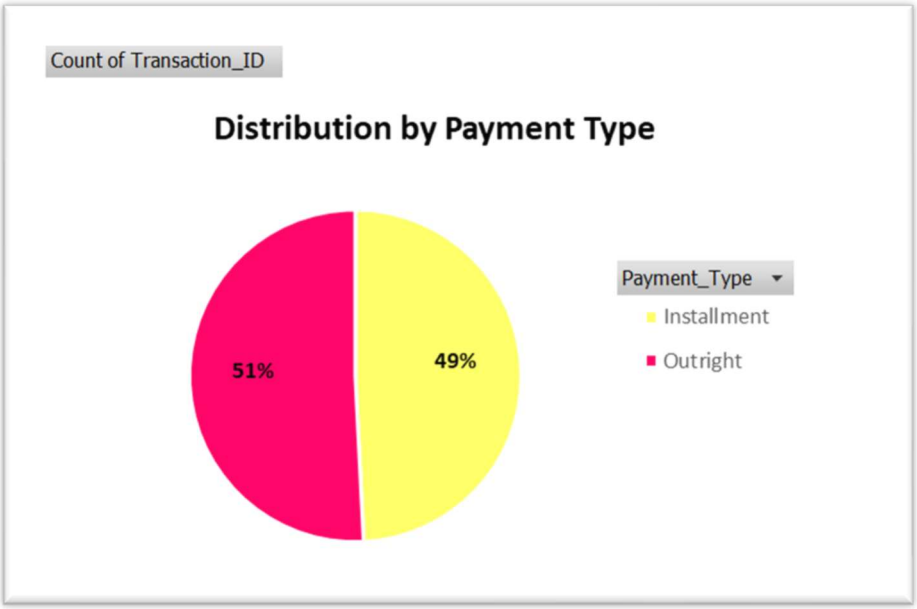
Property type	Sum of Revenue
Commercial Land	₦2,667,382,613
Mixed-use Land	₦2,508,161,112
Residential Land	₦2,562,743,831
Grand Total	₦7,738,287,556

Commercial Land generated the highest number of revenue

Payment Type Analysis

Payment Type	Transactions	Revenue
Installment	246	₦4,079,127,415
Outright	254	₦3,659,160,141

Outright payments account for 254 transactions while Installment accounts for 246. Interestingly, Installment generates higher total revenue than outright, suggesting that installment plans attract higher-value deals.



Average Land Size by Region

Region	Average Land Size (SQM)
North East	620.02
South East	616.91
South West	608.28
North West	604.68
South South	602.39
North Central	564.25

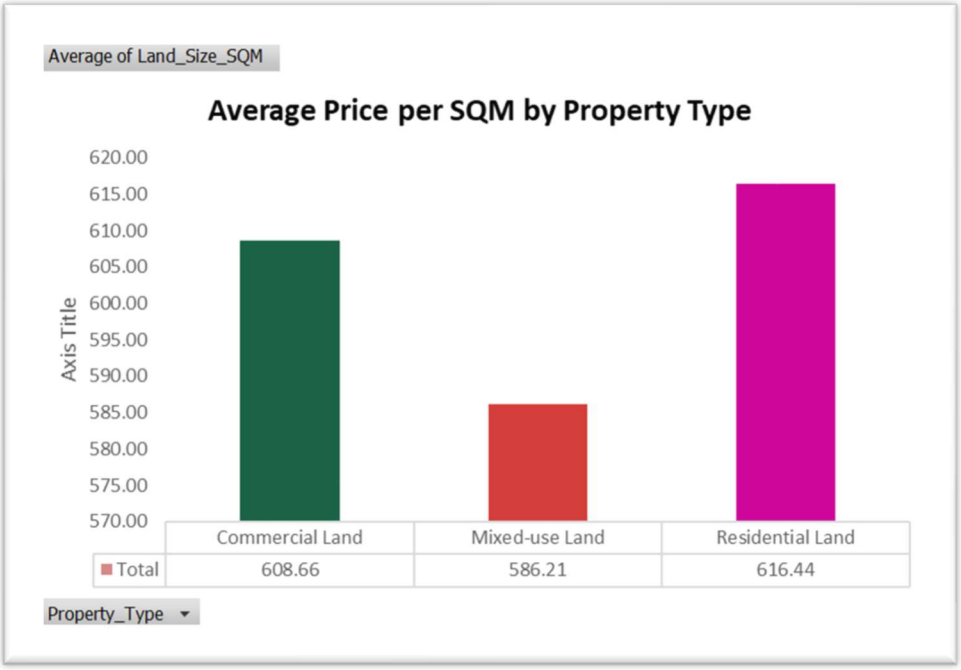
North East has the highest average land size, while North Central has the lowest.

Average Price per SQM by Property Type

Property Type	Average Price/SQM
Mixed-use Land	₦27,178
Residential Land	₦26,968

Commercial Land	₦26,638

Mixed-use Land had the highest average price per square meter, reflecting its flexibility and dual-use potential.



Initial Deposit by Company

Company	Total Initial Deposit
GreenField Estates	₦1,230,113,895
MetroLand Developers	₦1,145,531,742
GoldenPlots Nigeria	₦1,013,828,250
UrbanEdge Realty	₦1,022,999,733
PrimeLand Ltd	₦892,720,457

GreenField Estates collected the highest total initial deposits. This metric is important for cash flow assessment of each firm.

MULTI-DIMENSIONAL ANALYSIS

Revenue across Regions and Property types

Cross tabulating region and property type reveals that Commercial Land dominates revenue in South South and South West. Residential Land leads in North East and North Central. Mixed-use Land performs strongly in South East and North West.

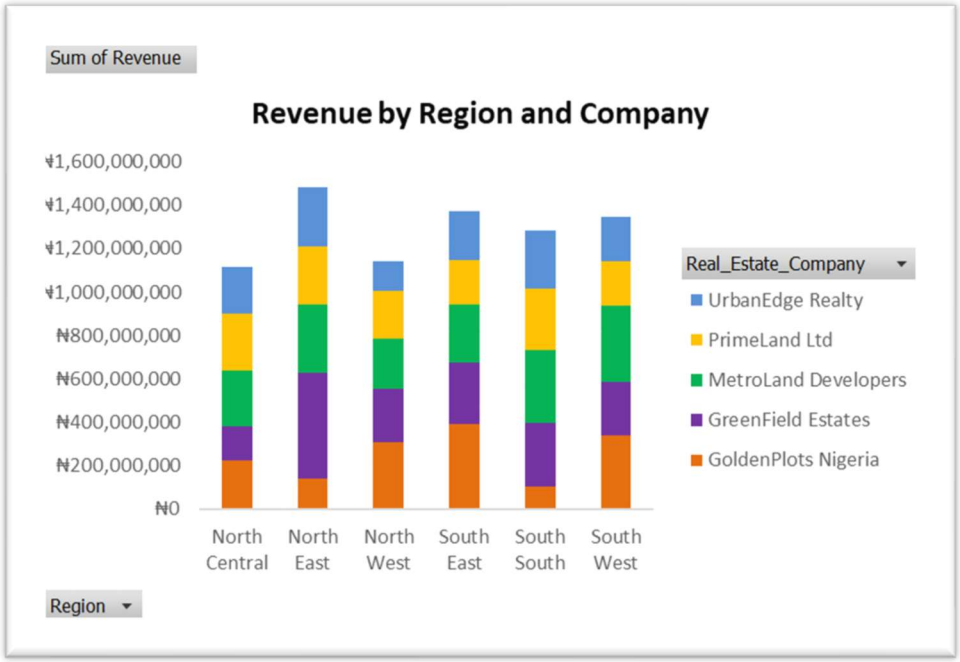
Region	Commercial Land	Mixed-use Land	Residential Land
North Central	₦240,681,820	₦395,130,018	₦479,711,171
North East	₦455,368,183	₦467,438,950	₦557,013,037
North West	₦396,977,192	₦416,112,863	₦326,602,427
South East	₦417,451,151	₦491,042,530	₦465,175,998
South South	₦615,337,674	₦292,880,811	₦377,185,216
South West	₦541,566,593	₦445,555,940	₦357,055,981

Best-performing property type by region:

- North Central → Residential Land (₦479,711,171)
- North East → Residential Land (₦557,013,037)
- North West → Mixed-use Land (₦416,112,863)
- South East → Mixed-use Land (₦491,042,530)
- South South → Commercial Land (₦615,337,674)
- South West → Commercial Land (₦541,566,593)

This demonstrates that property demand differs across regions

Company Contribution within Regions



Company strength varies by region. GreenField Estates is particularly strong in the North East (₦488,374,880). MetroLand Developers leads in South West and South South. GoldenPlots Nigeria shows relative strength in South East and North West. This regional specialization suggests opportunities for targeted expansion or partnership strategies.

Customer types across Property types

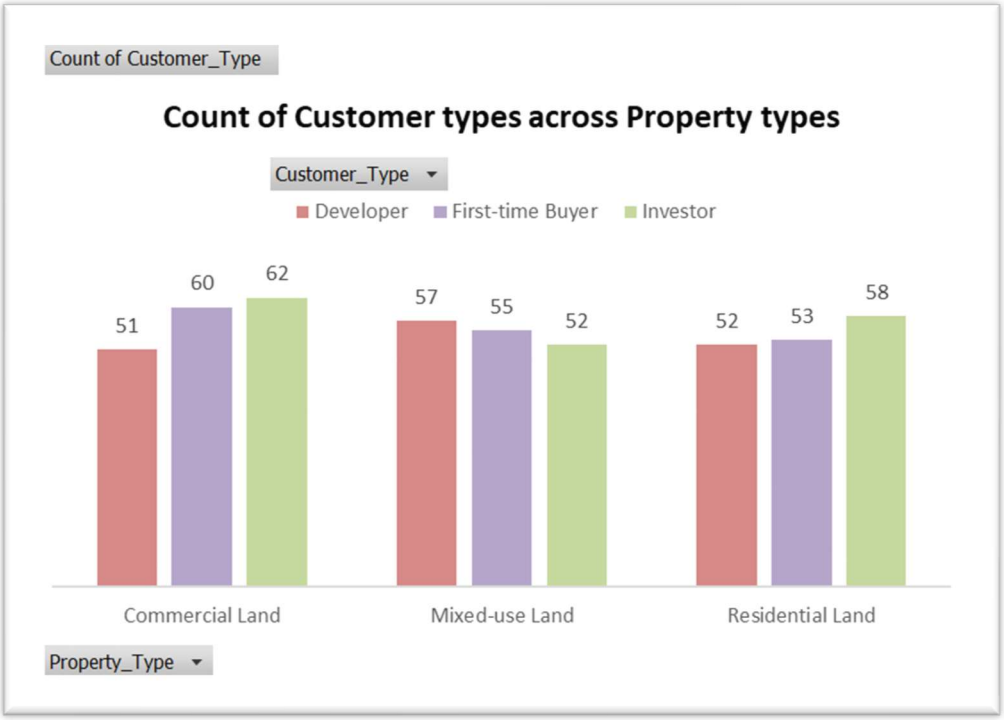
Property type	Developer	First-time Buyer	Investor
Commercial Land	51	60	62
Mixed-use Land	57	55	52
Residential Land	52	53	58
Grand Total	160	168	172

Property type	Developer	First-time Buyer	Investor

Commercial Land	₦762,710,238	₦1,068,028,086	₦836,644,289
Mixed-use Land	₦863,768,729	₦876,563,985	₦767,828,398
Residential Land	₦762,978,972	₦835,894,613	₦963,870,245
Grand Total	₦2,389,457,940	₦2,780,486,684	₦2,568,342,932

First-time buyers generated the highest revenue for Commercial Land and Mixed-used Land, while Investors generated particularly strong revenue from Residential Land.

The revenue relationship indicates that customer preferences differ by property type, suggesting that marketing strategies can be customized according to both customer segment and property category.



Area type effects on Revenue and Land size

Area type	Sum of Land_Size_SQM	Sum of Revenue
Rural	100792	₦2,467,532,677
Semi-Urban	95706	₦2,488,637,347
Urban	105418	₦2,782,117,532
Grand Total	301916	₦7,738,287,556

Urban transactions generated the highest total revenue with 105,418 SQM of land sold, while semi-urban follows in revenue but have the least of SQM of land sold. While Rural has the second in highest SQM of land sold but generated the lowest revenue.

PRICE AND LAND ANALYSIS

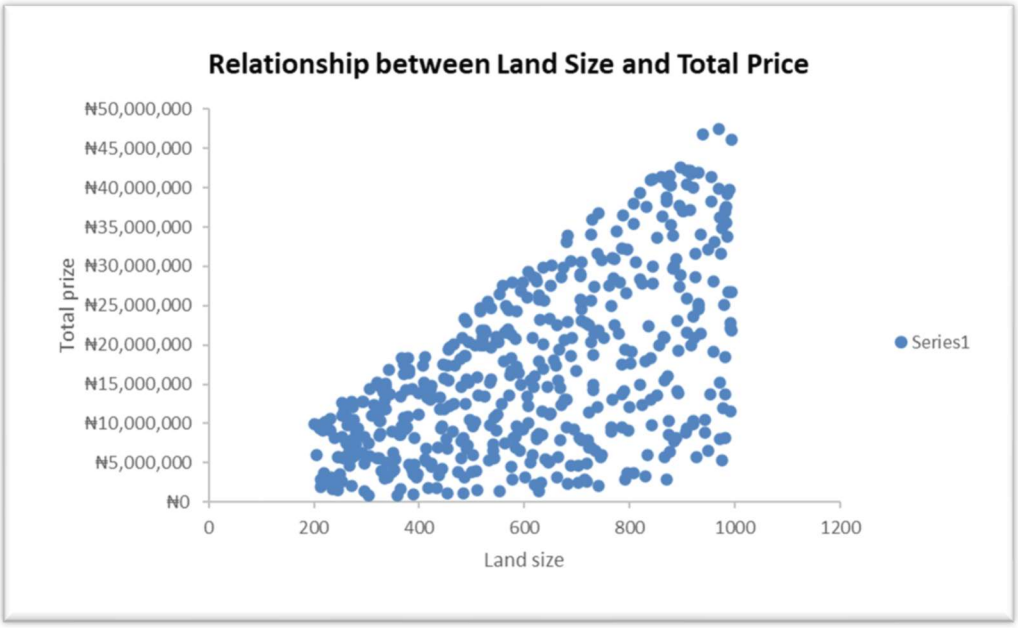
Relationship between land size and total price using scatter plot

There is a moderate positive relationship, larger plots generally tend to have higher total prices, although land size alone does not determine the final price.

The scatter plot therefore shows an upward tendency, but with considerable variation between individual transactions.

When land sizes are grouped into categories, a clear positive relationship emerges:

Land size	Sum of Revenue
200-464	₦1,315,007,993
465-729	₦2,739,752,783
730-994	₦3,683,526,780
Grand Total	₦7,738,287,556



Region with Highest Average Price per SQM

Region	Average of Price_per_SQM
North Central	₦26,417
North East	₦25,920
North West	₦28,176
South East	₦24,452
South South	₦31,194
South West	₦26,736

South South records the highest average price per square meter at ₦31,194, followed by North West (₦28,176). South East has the lowest average (₦24,452).

Land size distribution Across Region

Region	Sum of Land_Size_SQM	Average of Land_Size_SQM
North Central	45140	564.25
North East	60762	620.02
North West	43537	604.68
South East	58606	616.91
South South	42167	602.39
South West	51704	608.28

North East has the highest total land size and also the highest average.

Property type with the highest average land size

Property type	Average of Land_Size_SQM
Commercial Land	608.66
Mixed-use Land	586.21
Residential Land	616.44

Residential Land had the highest average land size (616.44), followed by Commercial Land (609 SQM) and Mixed-use Land (586 SQM). Residential buyers tend to acquire larger plots, consistent with housing and estate development needs.

TIME-BASED ANALYSIS

Total transactions for each month

Month	Count of Transaction ID
January	35
February	40
March	41
April	30
May	38
June	35
July	48
August	49
September	52
October	48
November	42
December	42

Transaction counts peak in September, with the highest transactions (52), August (49), July (48) and October (48). The quietest month is April (30). Activity is generally stronger in the second half of the year.

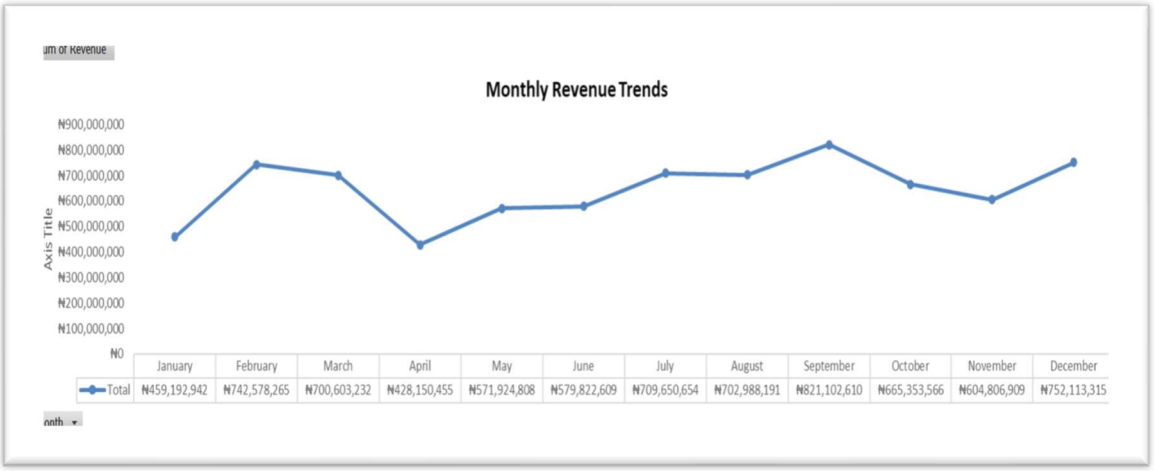
The monthly revenue trends

Month	Sum of Revenue
January	₦459,192,942
February	₦742,578,265
March	₦700,603,232

April	₦428,150,455
May	₦571,924,808
June	₦579,822,609
July	₦709,650,654
August	₦702,988,191
September	₦821,102,610
October	₦665,353,566
November	₦604,806,909
December	₦752,113,315

September recorded the highest number of transactions, with 52 transactions, and also generated the highest monthly revenue at approximately ₦821,102,610. April had the lowest transaction count and the lowest monthly revenue.

For Peak sales months, September, August, July, October has the peak sales month in that order.



Installment vs Outright Payments Over Time

Installment payments generated approximately ₦4,079,127,415, compared with ₦3 659,160,141 from outright payments. While they have total transactions of Installment (92) completed transactions, Outright (86) completed transactions.

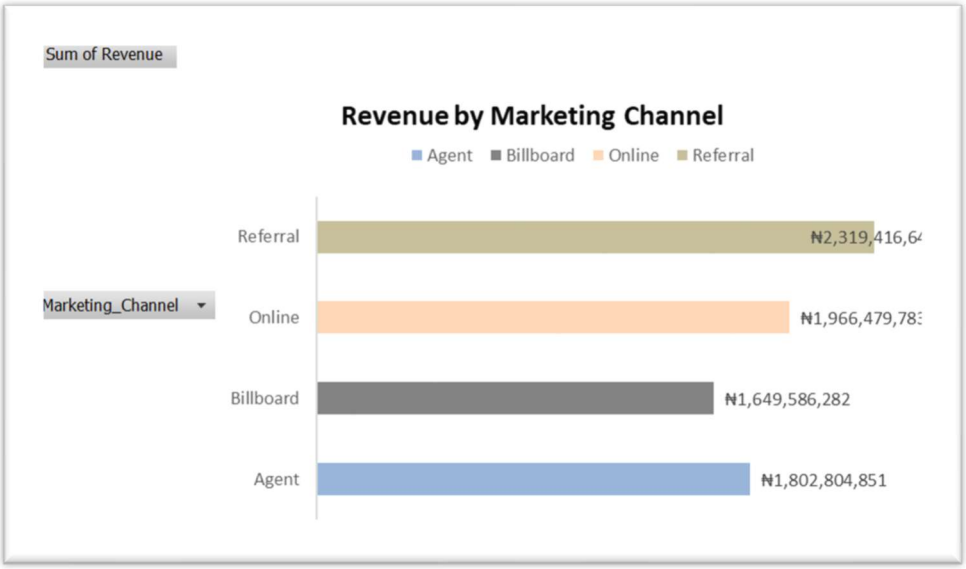
MARKETING CHANNEL AND ANALYSIS

Marketing Channel with highest Revenue

Marketing Channel	Count of Transaction ID	Sum of Revenue
Agent	117	₦1,802,804,851
Billboard	115	₦1,649,586,282
Online	138	₦1,966,479,783
Referral	130	₦2,319,416,640

Online generated the largest number of transactions, while Referral generated the highest revenue.

This distinction is important because the channel producing the most transactions is not necessarily the channel producing the most revenue.



Customer types across Marketing channel

Marketing Channel	Developer	First-time Buyer	Investor
Agent	37	38	42
Billboard	40	37	38
Online	46	50	42
Referral	37	43	50

Online marketing channel brought more customers than the others with a total of 138, and having First time buyer as the most transaction in that 138. Followed by Referral with 130, then Agent with 117 and Billboard with 115.

CONDITIONAL CALCULATIONS

Revenue Classification

Transactions were classified using the revenue thresholds in the workbook:

- High Value: Revenue ≥ ₦30 million
- Moderate Value: Revenue ≥ ₦10 million and below ₦30 million
- Low Value: Revenue below ₦10 million

Classification	Transactions
High Value	54
Moderate Value	251
Low Value	195

Most transactions fall into the Moderate Value category.

Land Size Classification

The dataset classified land sizes as:

- Small: below 465 SQM
- Medium: 465–729 SQM
- Large: 730 SQM and above

Category	Transactions
Medium	183
Large	161
Small	156

Medium sized plots were the most common.

ADVANCED CONDITIONAL ANALYSIS

Installment Transactions with Discount Above 5%

	Discount >5%	
Payment type	NO	YES
Installment	134	112
Outright	125	129
Grand Total	259	241

There were 112 installment transactions where the discount was greater than 5%.

High-Value Transactions in Urban Areas

There were 19 transactions classified as both High Value and Urban. These transactions represent an important segment for further investigation because they combine high transaction value with urban location.

Area type	Count of Transaction ID
Rural	**175**
High Value	16
Low Value	74
Moderate Value	85
Semi-Urban	**153**
High Value	19

Low Value	60
Moderate Value	74
Urban	**172**
High Value	19
Low Value	61
Moderate Value	92
Grand Total	500

PERFORMANCE ANALYSIS

Best performing Real Estate company

Real Estate Company	Sum of Revenue
GoldenPlots Nigeria	₦1,505,968,892
GreenField Estates	₦1,713,381,892
MetroLand Developers	₦1,757,314,263
PrimeLand Ltd	₦1,443,490,611
UrbanEdge Realty	₦1,318,131,897

MetroLand Developers leads with ₦1,757,314,263 in revenue and 114 transactions. GreenField Estates is a close second (₦1,713,381,892).

Region with highest Revenue

Region	Sum of Revenue

North Central	₦1,115,523,010
North East	₦1,479,820,169
North West	₦1,139,692,482
South East	₦1,373,669,679
South South	₦1,285,403,701
South West	₦1,344,178,514
Grand Total	₦7,738,287,556

North East has the highest.

Property type with highest revenue

Property type	Sum of Revenue
Commercial Land	₦2,667,382,613
Mixed-use Land	₦2,508,161,112
Residential Land	₦2,562,743,831
Grand Total	₦7,738,287,556

Commercial Land contributes the largest revenue, followed by Residential and Mixed-use.

Payment type with highest Revenue

Payment type	Sum of Revenue
Installment	₦4,079,127,415

Outright	₦3,659,160,141
Grand Total	₦7,738,287,556

Installment payments generate the highest revenue while Outright generates the lowest. Offering flexible payment plans appears to unlock higher overall sales value.

CUSTOMER BEHAVIOR ANALYSIS

Customer Type Purchasing the Most Land

Customer Type	Total Land Purchased
First-time Buyer	105,782 SQM
Investor	102,173 SQM
Developer	93,961 SQM

First-time buyers purchased the largest total amount of land

Customer Type Generating the Highest Revenue

Customer type	Sum of Revenue
Developer	₦2,389,457,940
First-time Buyer	₦2,780,486,684
Investor	₦2,568,342,932
Grand Total	₦7,738,287,556

First-time buyers generated the highest total revenue.

Customer Age and Spending

Customer type	Sum of Revenue
Developer	**₦2,389,457,940**
20-29	₦539,211,291
30-39	₦516,150,930
40-49	₦555,745,393
50-59	₦452,858,173
60-69	₦325,492,153
First-time Buyer	**₦2,780,486,684**
20-29	₦539,939,756
30-39	₦477,871,371
40-49	₦773,699,719
50-59	₦636,009,298
60-69	₦352,966,540
Investor	**₦2,568,342,932**
20-29	₦602,571,327
30-39	₦541,980,103
40-49	₦558,661,078
50-59	₦597,479,451
60-69	₦267,650,974

Revenue by broad age group shows that the 40–49 age group contributes the most revenue, followed by 50–59 and 20–29. Buyers aged 40–49 appear to be in their peak investment or home acquisition years. Therefore, age appears to be a strong predictor of transaction value.

Satisfaction by Customer Type

Customer Type	Average Satisfaction
Investor	3.16
First-time Buyer	3.06
Developer	2.93

Investors recorded the highest average satisfaction rating, while Developers recorded the lowest.

GROUPING, SORTING AND FILTERING

Top Five Transactions by Total Price

Transaction ID	Revenue
TRX-0177	₦40,322,314
TRX-0443	₦40,127,184
TRX-0233	₦38,828,756
TRX-0125	₦38,385,333
TRX-0302	₦37,924,611

These transactions represent the 5 highest value transactions based on Total Price.

OVERALL ANALYTICAL INTERPRETATION

- Which region generates the highest revenue?

The North East generated the highest total revenue at ₦1,479,820,169.

- Which property type contributes the most revenue?

Commercial Land generated the highest revenue at ₦2,667,382,613.

- Do larger land sizes generate higher revenue?

There is a moderate positive relationship between land size and revenue. Larger plots generally tend to generate higher revenue, although other factors such as location and price per SQM also influence revenue.

- Which company performs best based on revenue?

MetroLand Developers recorded the highest revenue ₦1,757,314,263, and also had the highest number of transactions.

- Which customer segment generates the most revenue?

First-time Buyers generated the highest revenue at ₦2,780,486,684 and also purchased the largest total amount of land.

- Which marketing channel is most effective?

The answer depends on the performance measure, Online generated the most transactions with 138 transactions, while Referral generated the highest revenue ₦2,319,416,640. Therefore, transaction volume and revenue produce different conclusions about channel performance.

- Does installment payment increase sales?

Installment transactions generated higher total revenue than outright, but Outright has higher transactions than Installment, so No, installment does not increase sales but helps generate higher income.

KEY BUSINESS INSIGHTS

The analysis produced several important findings:

1. The North East has the highest revenue region.

2. Commercial Land generated the highest total revenue.

3. MetroLand Developers recorded the highest company revenue.

4. First-time Buyers generated the highest customer revenue and purchased the largest amount of land.

5. Referral marketing generated the highest revenue, while Online generated the highest number of transactions.

6. September recorded the highest monthly transaction volume and revenue.

7. South South had the highest average price per square meter.

8. Residential Land had the highest average land size.

9. Larger land sizes showed a moderate positive relationship with revenue.

10. Customer age showed a relationship with spending.

11. Installment transactions generated more total revenue than outright transactions, but this should not be interpreted as causal evidence because Outright has more transaction sales.

12. Urban areas generated the highest total revenue and had a higher average transaction price than rural areas.

RECOMMENDATIONS

Based on the descriptive findings, the following business actions can be considered:

- Regional strategy: Investigate the factors behind the strong performance of the North East and identify whether the pattern is driven by transaction volume, property mix, pricing, or customer composition.
- Property strategy: Commercial Land generated the highest revenue, so its demand and pricing characteristics should be examined further.
- Customer strategy: First-time Buyers contributed the highest revenue and land volume, making this segment important for customer acquisition and retention analysis.
- Marketing strategy: Referral generated the highest revenue, while Online generated the highest transaction count. Both channels should therefore be evaluated using separate volume and revenue KPIs(Key Performance Indicator).

- Payment strategy: Installment payment should be analyzed further because it generated higher revenue in this dataset, particularly by comparing similar properties and customer segments.
- Pricing strategy: The relationship between land size, location, and price per SQM should be monitored when setting property prices.
- Dashboard monitoring: Management should use the interactive dashboard to monitor regional performance, property demand, customer behavior, payment patterns, and marketing performance.

CONCLUSION

The analysis of 500 real estate transactions provides a detailed view of sales performance and customer investment behavior across different regions and property categories in Nigeria.

The dataset shows that the North East generated the highest revenue, Commercial Land contributed the most revenue among property types, and MetroLand Developers recorded the highest company revenue. First-time Buyers were the largest contributor to customer revenue and total land purchased, while Referral marketing produced the highest revenue among marketing channels.

The analysis also found a moderate positive relationship between land size and transaction value, while customer age had positive relationship with spending. Monthly analysis showed September as the strongest month in terms of both transaction volume and revenue.